Taxonomies and tagging in MISP 2.5

Classify consistently, filter precisely, share deliberately

CIRCL / MISP Project

https://www.misp-project.org/

Reference snapshot: 29 October 2026

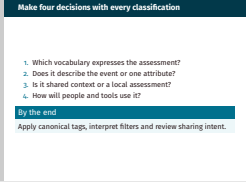


Make four decisions with every classification

1. Which vocabulary expresses the assessment?
2. Does it describe the event or one attribute?
3. Is it shared context or a local assessment?
4. How will people and tools use it?

By the end

Apply canonical tags, interpret filters and review sharing intent.



This is an operational course for the reviewed MISP 2.5.50 source. The capabilities shown are available in that snapshot; they were not all introduced in 2.5. Use the README's immutable sources to audit behavior. Interface labels and permissions vary by theme, role and configuration.

Four concepts that answer different questions

Concept	Purpose	Example
Taxonomy	Shared classification vocabulary.	Reliability, confidence, handling.
Tag	A label attached to data.	`tlp:amber`
Galaxy cluster	A described entity or concept with relationships.	A technique or threat actor.
Warninglist	Match values needing extra scrutiny.	Special-use IP ranges.

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Concept	Purpose	Example
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A custom free-text tag does not need a taxonomy, but it needs an agreed meaning to interoperate. Galaxy clusters are attached through tags yet carry richer structured knowledge. Warninglists compare observable values rather than assigning an analyst's classification. These concepts complement each other.

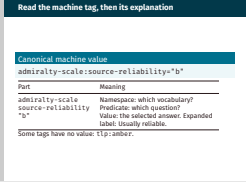
Read the machine tag, then its explanation

Canonical machine value

```
admiralty-scale:source-reliability="b"
```

Part	Meaning
`admiralty-scale`	Namespace: which vocabulary?
`source-reliability`	Predicate: which question?
`"b"`	Value: the selected answer. Expanded label: Usually reliable.

Some tags have no value: `tlp:amber`.



Namespace and predicate are required; a value is optional. The old example used an expanded description as the machine value. Copy the canonical tag supplied by the taxonomy rather than translating it, changing its case or substituting the expanded label. Keep quoted values in API JSON with proper JSON escaping.

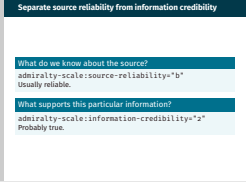
Separate source reliability from information credibility

What do we know about the source?

`admiralty-scale:source-reliability="b"`
Usually reliable.

What supports this particular information?

`admiralty-scale:information-credibility="2"`
Probably true.



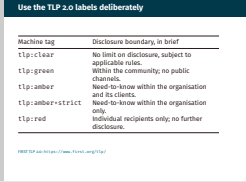
These are independent assessments, not an instruction to copy one score into the other. Other useful vocabularies include misp confidence-level and estimative-language. Numerical values are taxonomy metadata, not a universal calibrated probability or an automatic detection decision.

Choose and prepare the vocabulary

Use the TLP 2.0 labels deliberately

Machine tag	Disclosure boundary, in brief
`tlp:clear`	No limit on disclosure, subject to applicable rules.
`tlp:green`	Within the community; no public channels.
`tlp:amber`	Need-to-know within the organisation and its clients.
`tlp:amber+strict`	Need-to-know within the organisation only.
`tlp:red`	Individual recipients only; no further disclosure.

FIRST TLP 2.0: `https://www.first.org/tlp/`

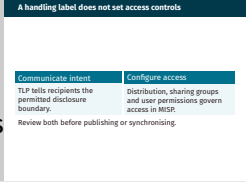


This is a teaching summary, not a replacement for FIRST's standard or the source's restrictions. The MISP vocabulary retains legacy and additional labels for compatibility; they are not all FIRST TLP 2.0 labels. Use CLEAR for new TLP 2.0 classifications; do not silently relabel existing information whose handling must first be reviewed.

A handling label does not set access controls

Communicate intent	Configure access
TLP tells recipients the permitted disclosure boundary.	Distribution, sharing groups and user permissions govern access in MISP.

Review both before publishing or synchronising.



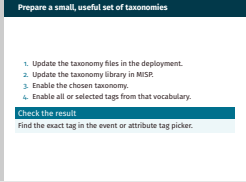
There is no universal one-to-one mapping between TLP labels and MISP distribution levels. A warning is not an enforcement guarantee. Tagging an event tlp:red does not automatically reduce its distribution or revoke copies already shared. Validate the actual recipients and any configured workflow or synchronisation rules.

Prepare a small, useful set of taxonomies

1. Update the taxonomy files in the deployment.
2. Update the taxonomy library in MISP.
3. Enable the chosen taxonomy.
4. Enable all or selected tags from that vocabulary.

Check the result

Find the exact tag in the event or attribute tag picker.



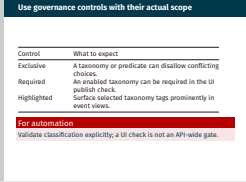
List Taxonomies is available at /taxonomies/index in the reviewed source; administrative actions require the corresponding permission. The library update reads app/files/taxonomies from the local deployment, not the latest GitHub revision on demand. Enabling the taxonomy and materialising its selectable tags are separate. Do not enable every vocabulary merely because it is available.

Use governance controls with their actual scope

Control	What to expect
Exclusive	A taxonomy or predicate can disallow conflicting choices.
Required	An enabled taxonomy can be required in the UI publish check.
Highlighted	Surface selected taxonomy tags prominently in event views.

For automation

Validate classification explicitly; a UI check is not an API-wide gate.

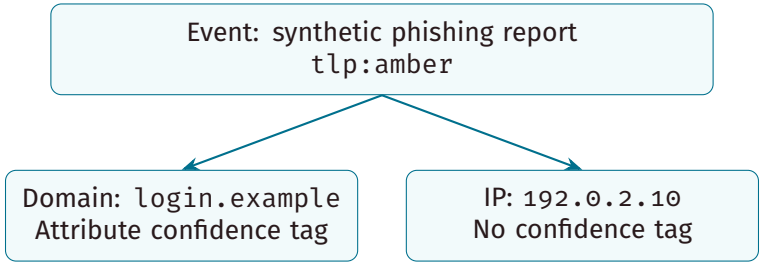


Required checks inspect event tags for the configured namespace; they do not prove the assessment is correct. EventsController's reviewed prepare-for-publish check is inside a non-REST branch. Do not advertise universal publishing enforcement. Exclusivity is checked in interactive tag-attachment paths and inconsistencies can be displayed. Recheck ingestion and automation paths; do not promise every path rejects conflicts.

Apply tags at the right scope

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Event context and attribute context are separate



Event tags are not copied onto each stored attribute.

Event context and attribute context are separate
Event: synthetic phishing report
tlp:amber
Domain: login.example
Attribute confidence tag
IP: 192.0.2.10
No confidence tag
Event tags are not copied onto each stored attribute.

The arrows show containment, not propagation. An object attribute is still an attribute when it is tagged. Filters and exports can include event context, which is different from an explicit attribute tag. Inspect the output envelope and options before concluding where a label was originally attached.

Choose shared context or a local assessment

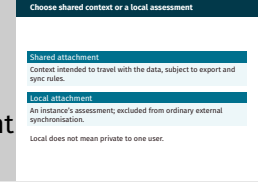
Shared attachment

Context intended to travel with the data, subject to export and sync rules.

Local attachment

An instance's assessment; excluded from ordinary external synchronisation.

Local does not mean private to one user.



The local flag belongs to the event-tag or attribute-tag association. The same label can be used with different attachment scope. Permissions and ownership determine who can apply each kind. Internal-server synchronisation can carry local tags, and local tags may appear in API exports unless excluded. This is not a secrecy boundary. The generic attachTagToObject route limits local tagging to the host organisation.

Three tag settings that are easy to confuse

Setting	Meaning
`local_only`	The tag can only be attached as a local tag.
`exportable`	Whether the tag is eligible for export; other rules still apply.
Organisation / user restriction	Who may apply the tag, not who may read the event.

Inspect tag settings and the attachment's local flag separately.

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Inspect tag settings and the attachment's local flag separately.

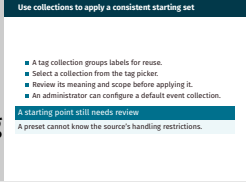
A Tag record's local-only and exportable fields do not replace the association's local flag. Administrative privileges can change assignment rules; UI and API routes enforce permissions. includeAllTags and excludeLocalTags affect exported visibility. Never use assignment restrictions as a substitute for distribution.

Use collections to apply a consistent starting set

- A tag collection groups labels for reuse.
- Select a collection from the tag picker.
- Review its meaning and scope before applying it.
- An administrator can configure a default event collection.

A starting point still needs review

A preset cannot know the source's handling restrictions.



Collections can include taxonomy tags and Galaxy cluster tags. Availability depends on collection sharing and member tag restrictions. The reviewed default setting is `MISP.default_event_tag_collection`. Applying a collection attaches its members; it is not a live policy link that automatically retags past events when the collection changes.

Keep the classification's meaning in context

- Event reports can also carry tags in the reviewed 2.5 source.
- A tag attachment can carry a relationship type.
- Use that annotation to explain how a label relates to its target.
- Keep the canonical taxonomy value unchanged.

Keep the classification's meaning in context
- Event reports can also carry tags in the reviewed 2.5 source.
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TagsController modifyTagRelationship supports event, attribute and event-report associations and checks modification permissions. Relationship metadata is distinct from a taxonomy predicate or a MISP object reference. Its availability in the interface and exports depends on the selected path; inspect what downstream tools consume. Analyst notes and opinions can record supporting reasoning.

An analyst’s tagging sequence

1. Read the evidence and the source’s handling instructions.
2. Choose the event or the specific attribute.
3. Select canonical tags and the intended attachment scope.
4. Review conflicts, distribution and the resulting export.

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Prefer a small agreed set over dozens of unmaintained labels. Record uncertainty rather than inventing confidence. A received event may allow local assessment but not global modification. A label alone does not publish data or activate a workflow.

Worked example: classify a synthetic report

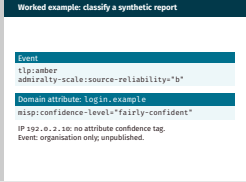
Event

```
tlp:amber
admiralty-scale:source-reliability="b"
```

Domain attribute: `login.example`

```
misp:confidence-level="fairly-confident"
```

IP `192.0.2.10`: no attribute confidence tag.
Event: organisation only; unpublished.



The full fixture is examples/tagged-event.json. All observables are reserved training values, not live indicators. This assessment does not equate source reliability with confidence. Do not publish the training event or enrich its values. The TLP label does not itself set distribution.

Find, process and review the data

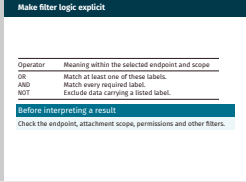
Find, process and review the data

Make filter logic explicit

Operator	Meaning within the selected endpoint and scope
OR	Match at least one of these labels.
AND	Match every required label.
NOT	Exclude data carrying a listed label.

Before interpreting a result

Check the endpoint, attachment scope, permissions and other filters.



MispAttribute dissectArgs accepts explicit OR, AND and NOT groups. Simple unprefixed lists form OR groups, not AND. Logic differs between event and attribute results; an event returned due to a matching attribute can contain other attributes. Prefer explicit groups over ambiguous string shorthand.

Search for attributes with both classifications

POST /attributes/restSearch

```
{
  "returnFormat": "json",
  "eventid": "7fb18198-5f8f-4fc3-8ea7-d2d791c2c0be",
  "tags": {
    "AND": [
      "tlp:amber",
      "misp:confidence-level=\"fairly-confident\""
    ]
  },
  "includeEventTags": true
}
```

The domain matches event handling *and* attribute confidence.

Search for attributes with both classifications
POST /attributes/restSearch
The domain matches event handling and attribute confidence.

Use the matching imported event UUID, not a global unbounded query. The eventid filter accepts an event ID or UUID in the reviewed source. In this attribute search, each positive tag may match the attribute or its event. The IP lacks the confidence tag and should not match. The README gives prerequisites. This request was source-checked, not executed against a live MISP instance.

A matching attribute can use event context

- Attribute tag searches can also match event tags.
- Request event tags explicitly when you need that context.
- Keep event and attribute provenance clear in consumers.
- Check returned values, not just the number of events.

In the example

`includeEventTags: true` adds context to the response; it does not change stored tags.

A matching attribute can use event context
Attribute tag searches can also match event tags.
Request event tags explicitly when you need that context.
Keep event and attribute provenance clear in consumers.
Check returned values, not just the number of events.
In the example
includeEventTags: true adds context to the response; it does not change stored tags.

Do not claim every endpoint has identical matching semantics. The example uses attributes/restSearch with an AND group and explicit event context. A local tag may influence searches on this instance; excluding local tags from output is not a universal instruction to ignore them during selection. Test the actual integration.

Tags help route work; configuration performs it

- Use labels such as `workflow:todo="review"` to coordinate work.
- Configure workflow conditions and actions for automation.
- Configure sync push/pull rules for the intended partner.
- Test a permitted and a rejected case before relying on the rule.

Tags help route work; configuration performs it
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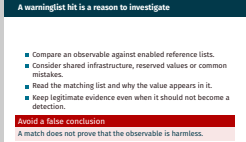
A workflow taxonomy tag is a label, not an executable workflow. A TLP tag does not automatically become a sync filter. Push/pull rules supplement distribution and authorisation; they do not expand access. Check event versus attribute filtering, local tags and transformations on the actual route. The dedicated workflows course covers configuration and execution.

A warninglist hit is a reason to investigate

- Compare an observable against enabled reference lists.
- Consider shared infrastructure, reserved values or common mistakes.
- Read the matching list and why the value appears in it.
- Keep legitimate evidence even when it should not become a detection.

Avoid a false conclusion

A match does not prove that the observable is harmless.



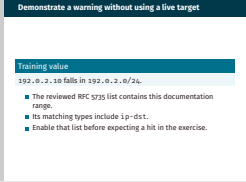
Warninglists have their own repository, update process and enabled state. A UI warning is not deletion or a taxonomy classification. Detection intent is separately represented by `to_ids`. The matching algorithm and applicable attribute types depend on the list.

Demonstrate a warning without using a live target

Training value

`192.0.2.10` falls in `192.0.2.0/24`.

- The reviewed RFC 5735 list contains this documentation range.
- Its matching types include `ip-dst`.
- Enable that list before expecting a hit in the exercise.



The current list is lists/rfc5735/list.json, a CIDR list. Other enabled lists may also match the same value. This proves a controlled training match, not maliciousness. Do not contact, enrich or scan the address.

Choose warning visibility or result suppression

`POST /attributes/restSearch`

Option	Behavior in the reviewed attribute search
`includeWarninglistHits`	Set to `true`: return matches with warning details.
`enforceWarninglist`	Set to `true`: omit matches from enabled applicable lists.

Compare the two supplied request files and inspect the actual response.

Choose warning visibility or result suppression
POST /attributes/restSearch
Option | Behavior in the reviewed attribute search
includeWarninglistHits | Set to true: return matches with warning details.
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Compare the two supplied request files and inspect the actual response.

examples/search-warnings.json and search-enforced.json use the same event and ip-dst scope. Enforcement removes matching result rows here; it does not delete stored data. Ingestion and other exports can implement different behavior. Enabled lists, changed filters and pagination can explain differences. Do not generalise to every API.

Diagnose the cause before creating another tag

Symptom	Check first
Tag missing in picker	Taxonomy, enabled terms, hidden tags and assignment restrictions.
Duplicate-looking labels	Canonical spelling, case and expanded display text.
Label missing downstream	Local/exportable flags, sync route and export options.
Unexpected search results	Endpoint, event versus attribute context and Boolean groups.

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Tag missing in picker	Taxonomy, enabled terms, hidden tags and assignment restrictions.
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Unexpected search results	Endpoint, event versus attribute context and Boolean groups.

Do not invent a free-text spelling to replace a missing picker entry from an existing taxonomy. Normalise or migrate only after reviewing semantics and downstream filters. Colour alone is not proof that labels have the same meaning.

Maintain and practise

A taxonomy definition starts with a clear question

```
{
  "namespace": "training-classification",
  "description": "Review state for a local exercise.",
  "version": 1,
  "predicates": [{
    "value": "review",
    "expanded": "Review state",
    "exclusive": true
  }]
}
```

This header is extended by values in the complete example file.

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}]
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This header is extended by values in the complete example file.

The header is valid JSON, not an excerpt with ellipses. examples/training-classification/machinetag.json contains the complete schema-valid definition. Exclusive is scoped to the review predicate. This is a teaching vocabulary; do not contribute it upstream unchanged. Search existing vocabularies such as workflow before creating a new one.

Values connect machine codes to readable labels

```
"values": [{
  "predicate": "review",
  "entry": [
    {"value": "draft", "expanded": "Draft"},
    {"value": "reviewed", "expanded": "Reviewed"}
  ]
}]
```

Resulting machine tag

```
training-classification:review="reviewed"
```

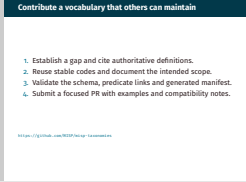

Values connect machine codes to readable labels
"values": [{ "predicate": "review", "entry": [{"value": "draft", "expanded": "Draft"}, {"value": "reviewed", "expanded": "Reviewed"}] }]
Resulting machine tag
training-classification:review="reviewed"

This is the values member inside the complete JSON object, not a standalone JSON document. The predicate must match a declaration. Stable codes support filters; expanded labels, descriptions and colours explain their meaning. Do not casually rename a published machine value.

Contribute a vocabulary that others can maintain

1. Establish a gap and cite authoritative definitions.
2. Reuse stable codes and document the intended scope.
3. Validate the schema, predicate links and generated manifest.
4. Submit a focused PR with examples and compatibility notes.

https://github.com/MISP/misp-taxonomies



Increase the version when editing an existing taxonomy, following the repository's workflow. Review references and applicable licensing. The JSON license offers CC0 or the stated redistribution license; do not make a blanket claim that all source vocabularies are public domain. Formatting and validation scripts may rewrite files and expect clean Git state.

Use the vocabulary outside a MISP instance

- Read the JSON vocabulary and its canonical values directly.
- Use PyTaxonomies to load and search taxonomy libraries.
- Validate against the current schema and repository checks.
- Record the library revision used by your consumer.

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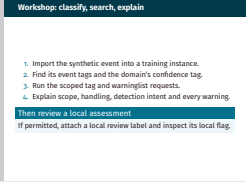
PyTaxonomies is separate; installing it does not enable a taxonomy in MISP. Its local-manifest CLI is exercised by taxonomy repository CI. The old 2016 output is removed rather than presented as current. The README provides offline schema validation for the sample.

Workshop: classify, search, explain

1. Import the synthetic event into a training instance.
2. Find its event tags and the domain's confidence tag.
3. Run the scoped tag and warninglist requests.
4. Explain scope, handling, detection intent and every warning.

Then review a local assessment

If permitted, attach a local review label and inspect its local flag.



Allow about 20 minutes after the presentation. The README gives prerequisites, expected outcomes and a debrief. The unpublished event has distribution zero and stays in training. Local tagging depends on role and host-organisation rules. Do not change sync configuration or publish to complete the exercise.

Before you share or automate

1. Are labels canonical, justified and attached at the right scope?
2. Do distribution and recipients respect handling instructions?
3. Have local context and warninglist hits been reviewed?
4. Does the real consumer interpret returned data correctly?

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Keep evidence and uncertainty visible. Separate assessment, access control, detection intent and export selection in human review and automation. Recheck after changing vocabularies, enabled lists, end-point options or partner routes. Validate the result.

References and the next practical steps

- Taxonomy reference and FIRST TLP 2.0.
- Taxonomy source and PyTaxonomies.
- Warninglists and MISP 2.5 source.
- This deck's `README.md`: exercise, build instructions, pinned evidence and validation limits.

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Continue with the Galaxy, workflows and data-model courses in this repository. The README maps technical claims to immutable evidence. Live links here lead to current project entry points; later checkouts may differ from this snapshot.